

MATHS ART: MANDALA ADVENTURES!



Discover Secret Math Patterns in Beautiful Art!



The Big Secret: Math is not just about numbers and equations on a blackboard—it lives inside art, beautiful festivals, and nature! When you draw shapes, clean lines, and perfectly matching designs, you are being a true mathematician!

Your Secret Missions!

- **Discover Mandalas:** Learn why "Mandala" means circle in the ancient Sanskrit language!
- **Be a Circle Tracer:** Find real objects to trace perfect circle rings of all different sizes.
- **Master Patterns:** Design repeating loops and dots that match perfectly all the way around.
- **Art Pride:** Create a colorful masterpiece you can show off at home!



Your Treasure Hunt Items:

Gather these fun tools from around your room or kitchen to start tracing:

- 2 Clean sheets of plain white paper
- 1 Pencil and 1 dark black sketch pen
- 1 Small coin or a round bangle
- 1 Medium bowl or a small bottle cap/lid
- 1 Big bowl or a large round plate
- 4 or more of your favorite colorful crayons or markers!

Magical Math Code Words

Shout these words out loud and match them with fun movements to remember your magical math formulas:

 **CIRCLE**

A perfectly round shape with no corners and no straight lines at all! Every spot on the outer edge is the exact same distance away from the middle.

CENTRE

The absolute perfect middle point of your circle. Everything we build today will grow outward from this one tiny hero dot!

PATTERN

Shapes, symbols, or colors that repeat in an exact, predictable order. Think of regular patterns on clothes, floor tiles, or natural layouts!

MANDALA

A beautiful circular design that expands outward from a single middle point. You see them every day in festive Indian Diwali Rangoli art, South Indian Kolam layouts, or majestic temple ceilings!

Let's Create Your Mandala! (6 Easy Steps)

Go slow and enjoy tracing. Remember, this is a secret mixture of math and art!

Step 1 The Center Dot: Lay your paper flat. Find the absolute middle of your page. Make a small, clear pencil dot right there. This is your center point where all your magic grows!

Step 2 The Inner Ring: Place your coin or round bangle flat on the paper directly over your center dot. Hold it steady with one finger, and carefully trace a pencil line around it. Lift it up to find your first small circle ring!

Step 3 The Middle Ring: Now choose your larger object (the medium bowl or lid). Center it carefully over your initial hero dot and trace all the way around its perimeter. Watch your paper circles look like ripples in clear water!

Step 4 The Petal Dance: Draw pretty flower petals filling up the empty space between your first small circle and your second larger circle. Make sure every single petal touches both line loops and repeats the same shape all the way around!

Step 5 Sparkle Details: Look at the empty spaces between your petals. Add a small decoration like a dot or a straight line in every single gap to make your artwork sparkle just like a traditional Diwali Rangoli or Kolam layout!

Step 6 The Color Pattern: Pick 2 or 3 of your favorite coloring markers. Create a regular repeating line pattern (like: Yellow, Blue, Yellow, Blue...). Color your matching petals and outer spaces to finish your masterpiece!

Example of a Repeating Color Pattern:



Red, Blue, Red, Blue, Red, Blue... That's a perfect pattern chain!

My Mandala Art & Math Journal

Part A: Draw what you learned today!

My Perfect Circle looks like this:



My Hero Center Dot goes here:



Part B: Record Your Color Secret Code

Draw or write the names of the colors you used for your masterpiece pattern:

 My Color Chain: _____ → _____ → _____ → _____

Part C: My Explorer Checklist

- ☐ I found the sweet center spot of my white paper [cite: 5].
- ☐ I successfully traced a small coin or bangle ring [cite: 5].
- ☐ I traced a bigger bowl ring safely without letting it slip [cite: 5].
- ☐ I added matching flower petals all the way around [cite: 5].
- ☐ I filled the gaps with tiny sparkle dots or lines [cite: 5].
- ☐ I colored everything following a steady pattern rule [cite: 5].

Home Pattern Detective Challenge

Your After-Class Mission:

Look closely around your house tonight! Can you spot a repeating shape or color pattern hiding somewhere? Check your mother's festival sarees, floor tiles, curtains, or porch designs. Draw the cool pattern you found in your notebook to become a certified **Mandala Maths Artist!** [cite: 5]